USED BIKE PREDICTION

1. Introduction

The automobile industry has experienced significant growth in the used vehicle market. Nowadays, many people prefer purchasing used bikes because they are more affordable compared to new vehicles. Used bikes provide a cost-effective transportation solution for students, employees, and daily commuters.

However, determining the correct resale value of a used bike is a challenging task because the price depends on various factors such as brand reputation, bike model, manufacturing year, engine capacity, kilometers driven, number of previous owners, fuel type, maintenance condition, and market demand.

Currently, used bike prices are mainly decided manually by dealers or owners based on their personal experience, current market trends, and comparison with similar bikes. This traditional method may result in incorrect pricing, which can cause financial loss to buyers and sellers.

To overcome these problems, an intelligent automated system is required to estimate the accurate resale value of used bikes. Machine Learning techniques can analyze previous bike sales data and identify patterns between different bike features and their selling prices.

The Used Bike Price Prediction System is a Machine Learning-based application that predicts the expected selling price of a used bike. The system collects important details about a bike from the user and applies a trained prediction model to calculate the estimated price.

This system helps buyers, sellers, and dealers make better decisions by providing a quick, reliable, and data-driven price estimation.

Objectives of the System:

• To develop an automated system for predicting used bike prices.

• To reduce errors in manual price estimation.

• To analyze historical bike sales data.

• To provide accurate price predictions using Machine Learning algorithms.

• To help users make better buying and selling decisions.

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2. Problem Definition

The major problem in the used bike market is estimating the actual market value of a bike. The resale price of a bike depends on multiple factors, making manual prediction difficult and unreliable.

Traditional pricing methods require users or dealers to compare different advertisements, check market conditions, and estimate depreciation manually. This process consumes more time and may not always provide accurate results.

The system needs to provide an automated solution that predicts bike prices based on different important attributes such as:

• Bike Brand
The brand value plays an important role in determining resale price. Popular brands usually maintain higher resale values.

• Bike Model
Different models have different market demands and price ranges.

• Manufacturing Year
The age of the bike affects depreciation. Newer bikes generally have higher resale values.

• Distance Traveled
The total kilometers driven indicate the usage level of the bike.

• **Number of Previous Owners**
A bike with fewer owners generally has a better resale value.

• Engine Capacity
Engine CC affects performance and pricing.

• Fuel Type
Petrol, electric, and other fuel types influence customer preference.

• Maintenance Condition
A well-maintained bike usually gets a higher price.

3. Existing System

In the existing system, used bike prices are estimated manually by dealers, sellers, or buyers. People usually depend on online advertisements, previous experience, and approximate market values.

The current process includes:

• Checking advertisements on online marketplaces.

• Comparing prices of similar bike models.

• Taking suggestions from dealers.

• Considering approximate depreciation values.

• Evaluating the physical condition of the bike manually.

Although this method is commonly used, it has several limitations.

Limitations of Existing System:

1. Manual Price Estimation

Price calculation is performed manually, which may lead to inaccurate predictions.

2. Time Consuming Process

Users need to search multiple websites and compare different bike prices.

3. Human Errors

Dealer experience and personal judgment may create incorrect price estimation.

4. Lack of Data Analysis

Traditional methods do not analyze large amounts of historical sales data.

5. Price Variation

Different sellers may provide different prices for the same bike model

6. No Automated Prediction

The existing system does not provide instant price prediction based on user inputs.

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4. Proposed System

The proposed system is a machine learning-based web application that predicts the selling price of a used bike based on the details entered by the user. The system uses a Regression algorithm (Random Forest Regressor) trained on historical used bike data to estimate an accurate market price.

The user enters information such as the bike brand, model, manufacturing year, kilometers driven, engine capacity (CC), fuel type, and owner details through a simple web interface. The application processes these inputs and sends them to the trained machine learning model. The model analyzes the input features and predicts the estimated selling price of the bike.

The application is developed using Python, Flask, HTML, CSS, and JavaScript. The machine learning model is trained using the Scikit-learn library and saved as a model file for future predictions. The web interface allows users to easily enter bike details and instantly view the predicted price.

Working of the Proposed System

1. User opens the Used Bike Price Prediction website.
2. User enters bike details such as Brand, Model, Year, Kilometers Driven, Owner, Engine CC, and Fuel Type.
3. The Flask application validates the input data.
4. The trained Random Forest Regression model processes the input.
5. The predicted selling price is generated.
6. The predicted bike price is displayed to the user on the result page.

Advantages of the Proposed System

- Provides quick and accurate bike price prediction.
- Reduces manual effort in estimating bike prices.
- User-friendly web interface.
- Helps buyers and sellers make informed decisions.
- Uses machine learning for improved prediction accuracy.
- Easy to maintain and update with new datasets.

The proposed system offers a reliable, efficient, and intelligent solution for estimating used bike prices by utilizing machine learning techniques and a simple web-based interface.